

Title: Prime Palindrome

Objective / Aim

To write a Java program to check whether a given number is both prime and palindrome.

System Requirements

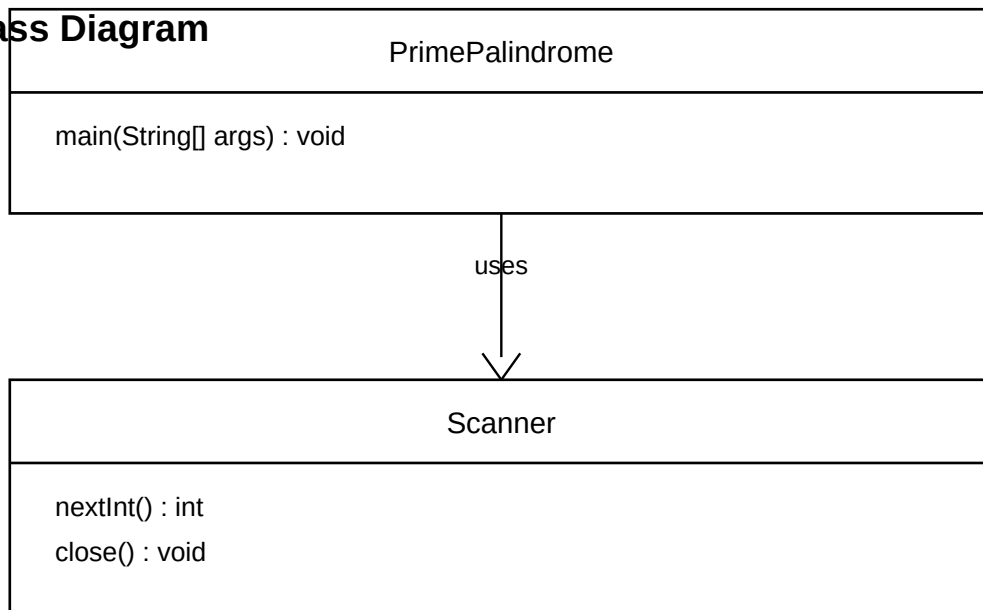
Hardware: Standard PC setup with Core i3/i5 processor and 4GB/8GB RAM.

Software: Windows/Linux/macOS, JDK 17/JDK 21, and Eclipse IDE or any Java IDE.

Algorithm / Logic Description

1. Import the Scanner class to read the number from the user.
2. Create a Scanner object and read the input number.
3. Initialize count to zero and check all numbers from 1 to the given number.
4. Increment count whenever the number is exactly divisible by the loop variable.
5. Reverse the given number using a while loop.
6. Compare the divisor count with 2 and compare the original number with its reverse.
7. If both conditions are true, display Prime Palindrome; otherwise display Not a Prime Palindrome.
8. Close the Scanner object.

UML Class Diagram



Source Code

```
import java.util.Scanner;

public class PrimePalindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
```

```
int num = sc.nextInt();

int count = 0;

for (int i = 1; i <= num; i++) {
    if (num % i == 0)
        count++;
}

int temp = num;
int rev = 0;

while (temp > 0) {
    int rem = temp % 10;
    rev = rev * 10 + rem;
    temp = temp / 10;
}

if (count == 2 && num == rev)
    System.out.println("Prime Palindrome");
else
    System.out.println("Not a Prime Palindrome");

sc.close();
}
```

Compilation & Execution Commands

```
javac PrimePalindrome.java
```

```
java PrimePalindrome
```

Output Screen

```
PS C:\Users\Student> javac PrimePalindrome.java
```

```
PS C:\Users\Student> java PrimePalindrome
```

```
Enter a number: 131
```

```
Prime Palindrome
```

```
PS C:\Users\Student>
```